

# 2026

# Industrial Immersion

# Poster Guidelines

Hi there,

On the next page you will find the poster template for your Industrial Immersion in 2026. It is pretty easy, especially if you follow our guidelines.

The purpose of the poster is to summarize your internship experience and showcase your internship results at the Skoltech Industry Day.

To be effective, the poster must be visually attractive and informative: include compelling visuals and clear, succinct prose.

If you have trouble, Industrial studies team can help: [internships@skoltech.ru](mailto:internships@skoltech.ru)

## General guidelines:

- It is critical to use the Microsoft Power Point to be sure the format of the poster is equal to the requirements for printing. Other software might mess up the suggested format of the poster when printed.
- Be sure to begin by checking the page size of your final poster size (A0 – 841x1189 mm)
- Do not adjust poster layout. Even if you are 100% sure that you know how to make it better that it is.
- Avoid jagged edges: left-justify text within text boxes — do not align text on the right or have it centered, whatever reason you find for this.
- Do not change the font and font color. Stick to the Arial, black. Check for this carefully when copying texts from other sources, formatting may be transferred.

## Content tips:

- Selectively incorporate charts, graphs, photos, maps that support the theme of your poster. Make sure they are referred to in the main body of the text, titled and captioned.
- Include your acknowledgements to those who helped and supervised in the Conclusions section, if any.

## When you are done:

- Edit your poster carefully for typos or grammatical mistakes before sending to supervisors' confirmation (use the print-preview function).
- Please, save your final version of the file in .ppt(x) format.

- Please, make sure that all information provided in the poster is provisionally verified and approved by a hosting company for publishing (NDA issues etc.).



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# Poster Guidelines:

# check out the sample

Insert the company's logo in here, make sure it fits well into the box — the poster will be printed on A0 format which is enough to represent the company properly. Please, indicate the company's official name in English in the text box above the logo

The headline is long, so the template adjusted the font size automatically

In case of a team project — list all students involved

The size of blank fields is more than enough to describe everything — avoid extending them

We recommend to divide your text into separate paragraphs in order to maintain clear and easy structure

Font size, style, color, line spacing, text indentation are the same in all segments, that's why this poster looks eye-catching and clean

Skoltech Industrial Immersion

2026

## Estimation of transport load capacity with complex surface geometry cargo

### Student(s)

Student Name, email, phone number  
Student Name, email, phone number  
Student Name, email, phone number  
Student Name, email, phone number

### Program

Program Title

### Skoltech supervisor

Skoltech supervisor Name

### Company supervisor

Company supervisor Name

In collaboration with  
Company Name



### Introduction

The aim of this project was to create software for accurate measurement of cargo volume with the intention of integrating it into production processes to improve the quality of work. These measures are necessary to improve the operational efficiency of the company, as well as the durability of the equipment.  
By implementing an automated system using Lidar technology, we sought to improve accuracy, reduce human error and optimize workflows.

### Objectives

- To develop a reliable system for accurate measurement of cargo volume.
- Implement an automated solution to optimize the workflow.
- To test the created system in real time.
- Ensure smooth integration of the new system with the company's existing management systems.

### Process & Methods

#### Installation development

The design of the installation was developed and agreed upon (Fig.1). The installation consists of:  
1. An aluminum housing  
2. A source of constant lighting  
3. Two 2D LIDARs positioned perpendicular to each other  
4. RGB camera, which is used to record and check a passing vehicle  
5. Radio tags for reading the model of the approaching vehicle and activating the software.

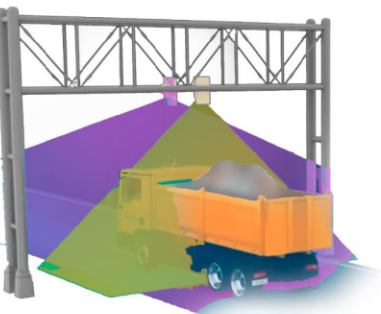
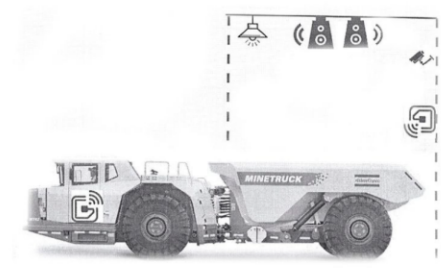


Figure 1: An illustration of equipment installation

Figure 2: An illustration of a cargo scan

#### Volume calculation system

Utilizing ROS 2 and Opened, a system for determining the volume of cargo was developed (Fig. 3) that can easily integrate after scanning a moving vehicle (Fig. 2).

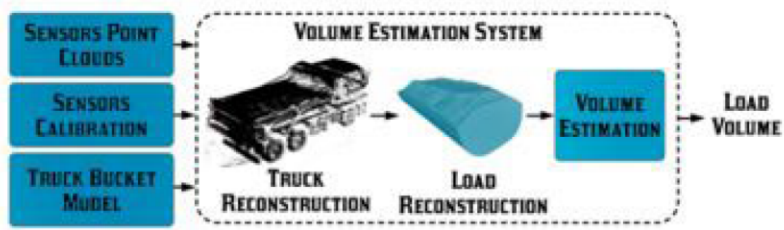


Figure 3: Overview of the system for estimating the load volume

The algorithm consists in combining two point clouds to obtain a three-dimensional point cloud after determining the route model and calibrating the sensors. After that, the point cloud corresponding to the real vehicle is restored in volume. Then, using segmentation, the points related to the cargo are determined and a three-dimensional model of the cargo is created. Next, the volume of the resulting shape is calculated. Finally, the actual volume of the cargo is determined. This algorithm of operation allows you to accurately and quickly get an accurate result.

### Results

The developed system accurately determines the cargo volume in trucks used in blasting operations through the use of dual 2D LIDARs and advanced data processing algorithms.

The system integrates seamlessly with existing production management frameworks, providing real-time volume data that enhances inventory management and logistics planning. The following key results were achieved:

- Accurate Volume Measurement: The system reliably measures cargo volume with high precision, minimizing errors associated with manual methods.
- Real-time Processing: Automated data collection and processing ensure continuous, real-time updates of cargo volume.
- Seamless Integration: The system's compatibility with existing production systems allows for immediate use and operational efficiency.
- Enhanced Inventory Management: Real-time volume data supports better resource allocation and logistical planning.

The system was thoroughly tested and validated, demonstrating its effectiveness and reliability in various operational conditions.

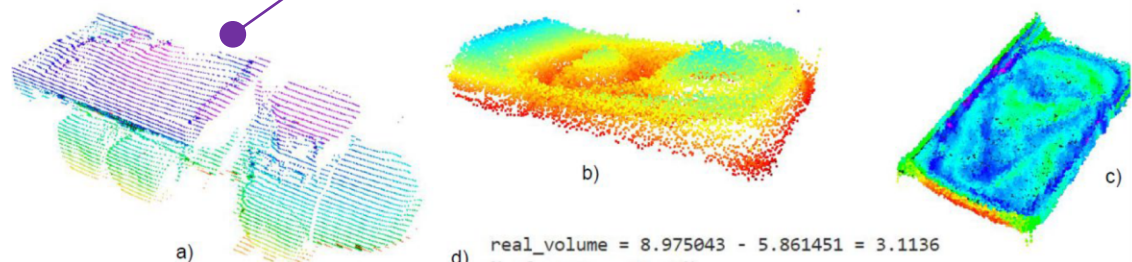


Figure 4: Illustration of the real work process

Figures 4a-d illustrate the system's capabilities:

- Figure 4a: Point cloud of the entire truck, showing the comprehensive scan of the vehicle and cargo.
- Figures 4b-c: Segmentation of the cargo from the point cloud, highlighting the isolation of the load from the rest of the truck.
- Figure 4d: Result of data processing using the developed program, demonstrating the final calculation and visualization of the cargo volume.

### Conclusions

This project presents an approach to cargo volume measurement in trucks by employing dual 2D LIDARs and advanced data processing techniques. The developed system enhances accuracy and efficiency through automated, real-time scanning and volume estimation. It integrates seamlessly with existing production management systems, improving inventory management and logistics planning. The solution has significant potential applications in various industrial settings where precise cargo volume measurement is critical. Future areas of work include the integration of this work in real time in real production.

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You can add picture(s) in sections **Process** and/or **Results**. Arrange them inside the provided space on your own

... and make sure your pictures are clearly captioned